

## EDUCATION

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- **Radboud University** Nijmegen, The Netherlands  
*PhD position in:* Calculus of Variations; *Supervisor:* Riccardo Cristoferi  
*September 2023 - August 2027*  
**Area of interest:** Calculus of variations, geometric measure theory, Gamma-convergence, material science and related fields.
- **Polytechnic University of Turin** Turin, Italy  
*Master degree in:* Engineering Mathematics  
*September 2021 - July 2023*  
**Thesis:** Anti-plane analysis of Francfort-Marigo model for quasi-static brittle fracture; *Supervisor:* Marco Morandotti.
- **KTH Royal Institute of Technology** Stockholm, Sweden  
*Exchange semester*  
*August 2022 - January 2023*
- **Polytechnic University of Turin** Turin, Italy  
*Bachelor degree in:* Mathematics for Engineering  
*September 2018 - July 2021*  
**Thesis:** Fractal dimension and Hausdorff measure; *Supervisor:* Luca Lussardi
- **L.S. Battaglini** Taranto, Italy  
*Scientific Diploma*  
*September 2013 - July 2018*

## PUBLICATIONS

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- **Low regularity potentials in heterogeneous Cahn–Hilliard functionals**  
*R. Cristoferi, J. Deutsch, L. Pignatelli*  
Arxiv: 2603.25369
- **Phase separation in multiply periodic materials with fine microstructures**  
*R. Cristoferi, L. Pignatelli*  
Arxiv: 2506.14462
- **Phase separation in multiply periodic materials with fine microstructures: a second regime**  
*L. Pignatelli*  
ongoing
- **Sharp interface dynamics in heterogeneous domains**  
*R. Cristoferi, L. Gancedi, L. Pignatelli*  
ongoing
- **Convergence of critical regime to sub- and super-critical regimes**  
*R. Cristoferi, L. Pignatelli*  
ongoing

## TALKS

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- **A Taxonomy of Separation: from Flory-Huggins to non-local gradients**  
*06-07-2026, Workshop Calculus of Variations NL*  
*Roermond*
- **Sharp interface limit of the Allen-Cahn equation with small periodic heterogeneities**  
*08-05-2026, Seminar on Calculus of Variations*  
*TU Wien, Vienna*
- **Why Soap Films are smarter than Calculus**  
*16-12-2025, PhD Colloquium*  
*Radboud University, Nijmegen*
- **Phase separation in multiply periodic materials with fine microstructures**  
*10-07-2025, Workshop Calculus of Variations NL*  
*Roermond*
- **Phase separation in heterogenous media**  
*28-05-2025, Seminar on Calculus of Variations*  
*TU Wien, Vienna*
- **Homogenization and phase separation in heterogeneous media**  
*20-03-2025, Seminar talk*  
*RWTH Aachen University, Aachen*
- **Gamma Mia - an introduction to Gamma convergence**  
*14-10-2024, PhD Colloquium*  
*Radboud University, Nijmegen*

- **Quasi-static brittle fracture: analysis of Francfort-Marigo model**

22-11-2023, Applied Analysis Seminar

Radboud University, Nijmegen

## POSTERS

- **Sharp interface limit of the Allen–Cahn equation with small periodic heterogeneities**

11-06-2025, Calculus of Variations and Applications to the Real World

INdAM Meeting, Cortona

- **Phase separation in heterogeneous media**

12-06-2025, Mathematical Analysis and Applications

Summer School, Como

- **Phase separation in multi-scale media**

11-04-2024, Mathematical Advances for Complex Materials with Microstructures

University of Würzburg, Würzburg

## RESEARCH VISITS

- **TU Wien, Vienna**

Collaboration with Jakob Deutsch

04-05-2026 - 08-05-2026

- **TU Wien, Vienna**

Collaboration with Jakob Deutsch

26-05-2025 - 30-05-2025

- **RWTH Aachen University, Aachen**

Collaboration with Likhit Ganedi

17-03-2025 - 20-03-2025

## PROJECTS

- **Reading Group organizer**

Series of meetings for the CalcVarNL community - Main topics: Gradient Flows, Optimal Transport, Stochastic Analysis.

The Netherlands, *September 2023 - ongoing*

- **Modeling Camp organizer**

Collaboration with dutch companies, where problems from the industry are modeled mathematically and numerically.

Radboud University, *February 2024, February 2025*

- **Head of YSC committee**

Young Scientist Council of the Institute for Mathematics, Astrophysics and Particle Physics.

Radboud University, *March 2024 - ongoing*

## EVENTS ATTENDED

- **Workshop Calculus of Variations NL**

06-07-2026 - 10-07-2026

Roermond, The Netherlands

- **Calculus of Variations and Applications to the Real World**

08-06-2026 - 12-06-2026

INdAM Meeting

Cortona, Italy

- **Modern Methods in Calculus of Variations**

23-02-2026 - 27-02-2026

Hausdorff Center for Mathematics

Bonn, Germany

- **Workshop Calculus of Variations NL**

07-07-2025 - 11-07-2025

Roermond, The Netherlands

- **Mathematical Analysis and Applications**

09-06-2025 - 13-06-2025

Summer School

Como, Italy

- **Analysis of Partial Differential Equations**

16-09-2024 - 19-09-2024

Czech Academy of Sciences

Prague, Czech Republic

- **Workshop Calculus of Variations NL**

08-07-2024 - 12-07-2024

Schiermonnikoog, The Netherlands

- **Vito Volterra Meeting in Calculus of Variations 2024**

24-06-2024 - 28-06-2024

Sapienza University of Rome

Rome, Italy

- **Mathematical Advances for Complex Materials with Microstructures**

08-04-2024 - 12-04-2024

University of Würzburg

Würzburg, Germany

- **Variational models for material failure**

20-09-2023 - 22-09-2023

FAU Erlangen-Nürnberg

Erlangen, Germany

- **Analysis of Partial Differential Equations**

18-09-2023 - 20-09-2023

KU Eichstätt-Ingolstadt

Eichstätt, Germany

- **Beyond Elasticity: Advances and Research Challenges**

16-05-2022 - 20-05-2022

CIRM

Luminy, France

## TEACHING

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- **Numerical Methods for PDEs**  
TA for the bachelor course Numerical Methods for PDEs  
*February 2026 - June 2026*  
prof. V. Nikolić
- **Numerical Methods**  
TA for the bachelor course Numerical Methods  
*September 2025 - January 2026*  
prof. L. Scarabosio
- **Calculus of Variations**  
TA for the MasterMath course Calculus of Variations  
*February 2025 - June 2025*  
prof. A. Bach, prof. R. Cristoferi
- **Multivariable Analysis**  
TA for the bachelor course Multivariable Analysis  
*September 2024 - January 2025*  
prof. R. Cristoferi
- **Numerical Methods for PDEs**  
TA for the bachelor course Numerical Methods for PDEs  
*February 2024 - June 2024*  
prof. V. Nikolić
- **Analysis 2**  
TA for the bachelor course Analysis 2  
*September 2023 - January 2024*  
prof. R. Cristoferi

## SKILLS

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- **Languages:** Italian (native), English (fluent), Dutch (basic), Spanish (intermediate)
- **Coding:** C++, Python, Matlab,  $\text{\LaTeX}$ – all advanced

## ACADEMIC REFEREES

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- **Riccardo Cristoferi:** Radboud University, cristoferi@science.ru.nl
- **Marco Morandotti:** Polytechnic University of Turin, marco.morandotti@polito.it

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Luca Pignatelli  
Radboud University, Nijmegen  
29-06-2026